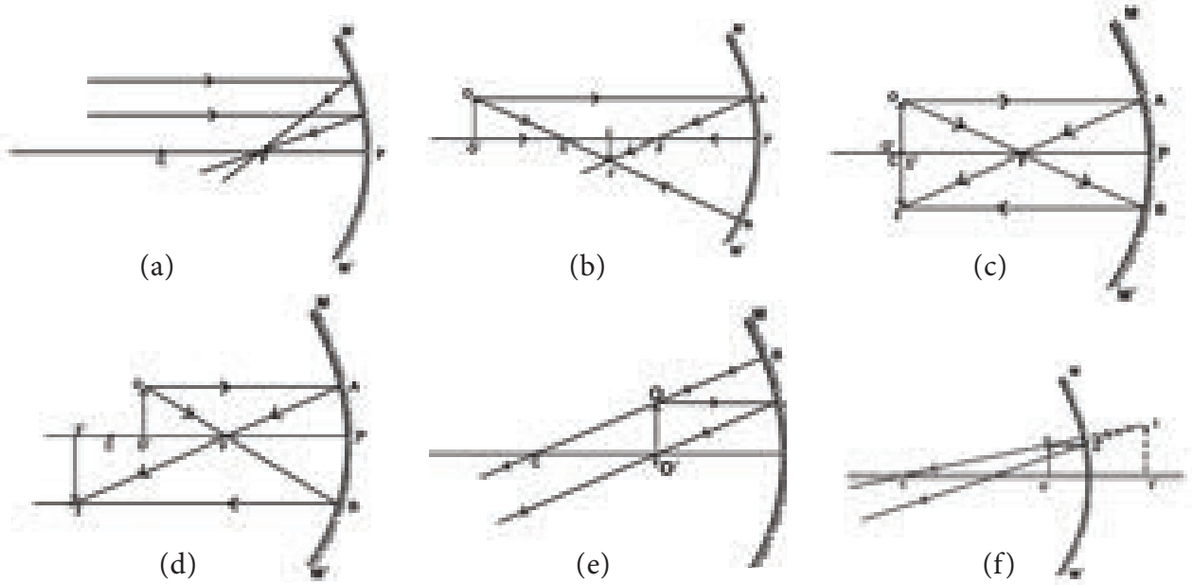


Figure 6.8 Ray diagram for the images formed by concave mirror

6.4.2 Sign convention for measurement of distances

We follow a set of sign conventions called the cartesian sign convention to measure distances in ray diagram. In this convention, the pole (P) of the mirror is taken as the origin. The principal axis is taken as the X-axis of the coordinate system (Figure 6.9).

- The object is always placed on the left side of the mirror.
- All distances are measured from the pole of the mirror.
- Distances measured in the direction of incident light are taken as positive and those measured in the opposite direction are taken as negative.
- All distances measured perpendicular to and above the principal axis are considered to be positive.
- All distances measured perpendicular to and below the principal axis are considered to be negative.

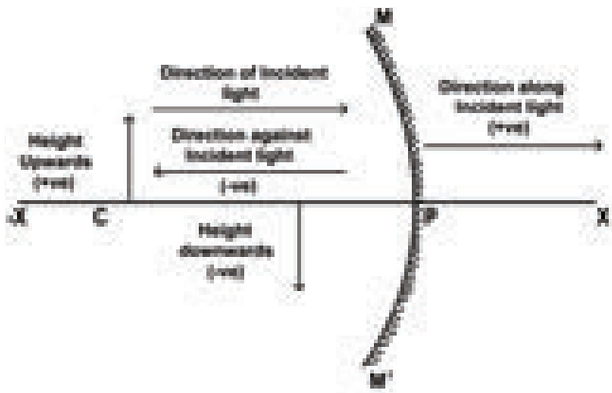


Figure 6.9 Sign convention for spherical mirrors

Table 6.1 Sign convention for spherical mirrors.

Type of mirror	u	v		f	R	Height of the object	Height of the image	
		real	virtual				real	virtual
Concave mirror	-	-	+	-	-	+	-	+
Convex mirror	-	No real image	+	+	+	+	No real image	+

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